

Outline

- Computer Memory
- Types of Computer Memory
- Memory Capacity
- Memory Hierarchies

Computer Memory

- Computer memory is any physical device, used to store data, information or instruction temporarily or permanently.
- It is the collection of storage units that stores binary information in the form of bits.
- The memory block is split into a small number of components, called cells.
- Each cell has a unique address to store the data in memory,

Memory components of a computer system can be divided into three categories :

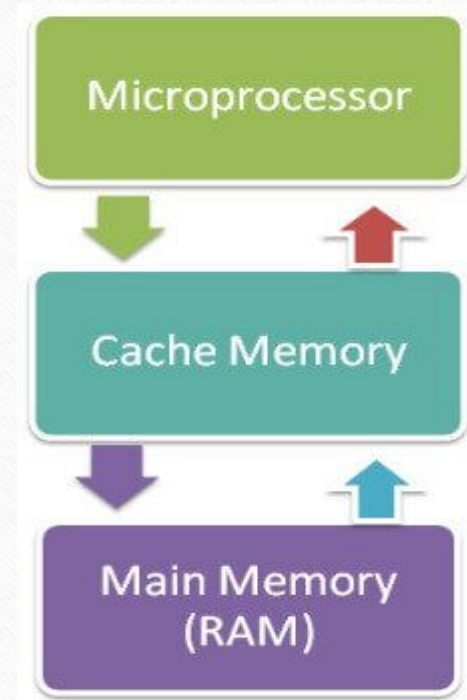
- Internal Processor Memory (or built-in memory)
- Primary memory or Main Memory
- Secondary memory

Internal Processor Memory

- It is not a part of the main memory and is located in the CPU in the form of registers, which are the smallest data holding elements.
- A register temporarily holds frequently used data, instructions, and memory address that are to be used by CPU.
- They hold instructions that are currently processed by the CPU.
- Some of the widely used Registers include Accumulator or AC, Data Register or DR, the Address Register or AR, Program Counter (PC), I/O Address Register, and more.
- Some microprocessors also employ another type of built in memory called cache memory.

Cache Memory

- Cache memory is a high-speed memory, which is small in size but faster than the main memory (RAM).
- The CPU can access it more quickly than the primary memory.
- So, it is used to synchronize with high-speed CPU and to improve its performance.



How does Cache Memory Work?

- Cache memory temporarily stores information, data and programs that are commonly used by the CPU.
- When data is required, the CPU will automatically turn to cache memory in search of faster data access.
- When data is found in cache memory, this is called a **cache hit**.
- A cache hit enables the processor to retrieve data quickly, making your overall system more efficient.
- If the data is not found in the cache (a cache miss), the CPU fetches the data from RAM and stores a copy in the cache for future use.
 - This process is slower than a cache hit, but it ensures that frequently accessed data is available in the cache for faster access later.

Primary Memory

Primary Memory is of two types: RAM and ROM.

RAM (Volatile Memory) :

- It is a volatile memory.
- It means it does not store data or instructions permanently.
- When you switch on the computer the data and instructions from the hard disk are stored in RAM.
- CPU utilizes this data to perform the required tasks.
- As soon as you shut down the computer the RAM loses all the data.



Primary Memory – Types of RAM

Types of RAM:

Integrated RAM chips can be of two types:

- Static RAM (SRAM):
- Dynamic RAM (DRAM):

DRAM (Dynamic Random Access Memory)

- Data is stored in capacitors.
- Capacitors that store data in DRAM gradually discharge energy, no energy means the data has been lost.
- So, a periodic refresh of power is required in order to function.
- DRAM is called dynamic as constant change or action(change is continuously happening) i.e. refreshing is needed to keep the data intact.
- It is used to implement main memory. It is typically used for main system memory (RAM)

SRAM (Static Random Access Memory)

- Data is stored in transistors and requires a constant power flow.
- Because of the continuous power, SRAM doesn't need to be refreshed to remember the data being stored.
- SRAM is called static as no change or action i.e. refreshing is not needed to keep the data intact.
- It is used in cache memories.

Primary Memory – Difference between SRAM and DRAM

SRAM	DRAM
Stands for Static Random Access Memory	Stands for Dynamic Random Access Memory
It does not need to be refreshed repeatedly.	It needs to be refreshed continuously or it will lose the data
High speed RAM	Relatively slower
Costly	Relatively low cost
Generally used for Cache	Used for main memory
Control complexity is less	Relatively high control complexity
Lower Access Time	Relatively higher access time

Thank You

